



System Acceptance Review

BRACU Mongol-Tori

BRAC University

University Rover Challenge - 2023



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1. Introduction: Humans are inquisitive to know the unknown, and the planet adjacent to us is no different. To learn more about it, the students of BRAC University set out on an exciting expedition with BRACU Mongol-Tori, a new generation Mars rover equipped to work alongside astronauts and conduct scientific research. Following the requirements of URC 2023, we are introducing “Mongol-Tori Encephalon” with an upgraded version of the Artificial Intelligence-powered subsystems. This year we completely redesigned the science and electrical systems with significant modifications to the autonomous, network, control, and mechanical systems to give the rover a fresh look. Minor changes include updating the claw, wheel, chassis, and EPS system. This report evaluates the Core Rover Systems, Approach to Competition Tasks, Testing and Operations, Team Budget, Gantt Chart, and Science Task in line with the standards of URC 2023.

2. Core Rover Systems: Using our previous experiences of URC and the IRC 2023- competition feedback, with some trial and error, we have brought changes to our entire rover system, with significant changes in the mechanical setup, electronics, and network system.

2.1 Mechanical: This year, we made changes to our rover design in the mechanical system after learning from previous experiences. We will use our four-wheeler setup with slight modifications to our rover body and wheels. A new claw design is already under development, as shown in **Figure 1(d)**.



Figure- 1(a) Full body rover



1(b) Modified extender



1(c) New Wheel



1(d) Adaptive Claw

2.1.1 Chassis and Suspension: The H-shaped ladder chassis of our rover is 0.7m long and 0.55m wide to house the circuit box and arm. Triangulation in the chassis prevents deformation in the chassis as the rover moves through rocky terrain. The heavy load is transmitted as tension by the heavy space frame and compression load along the length of the strut. We have used a custom rocker-bogie mechanism in our four-wheeler rover, where the two bogies are connected to the chassis. A U-shaped differential bar provides extra support for the rear part of the system. The two universal joints in the differential bars magnify the rotational angle of bogies, where the bogies and feedback actuators can turn from 1.2m to 0.76m. The new extender suspension is lightweight and swift, powered with a 65 RPM power window motor ensuring stability during vertical drops. The extender prevents flipping during steep drops by increasing the length of its base. A tilted wheel mechanism is added which makes the rover have a 60-degree angle with the wheel to give it more stability in slow descents. **Figure 1(a)** shows our full working rover's design. **Figure 1(b)** shows the use of our modified extender.

2.1.2 Wheel: For the movement of our rover, we used stainless steel wheels with a diameter of 0.29m and a width of 0.112m. The design of the wheel along with the steel plate rim makes it stronger and able to absorb shocks and withstand vertical drops. Small holes in the wheel and the rim make it lighter with added grip to give a 360-degree rotation. In addition, the wheels have 2.5mm thick modified rubber grippers, increasing the traction, as shown in **Figure 1(c)**.

2.1.3 Arm: The rover has a robust manipulator arm with the 360-degree rotational ability and six degrees of freedom (base, shoulder, elbow, wrist) with an end effector with two degrees of freedom. The wrist has a chain and sprocket system, with a worm gear at the base of our arm that locks the arm in place and prevents backlash. Our feedback system is made with a potentiometer in every arm's joint and implements an inverse kinematics algorithm for more precision. We have used a three-finger claw system for better control, precision, and gripping ability. A two-finger adaptive claw as shown in **Figure 1(d)** is under development, which will give more precision and control. For collecting soil, we use an excavator claw that can rotate on its axis to break down rocks and soil. We are using Gazebo for the 3D module simulation of the arm. To get precise visualization of the arm, we are using Rviz in our GUI.

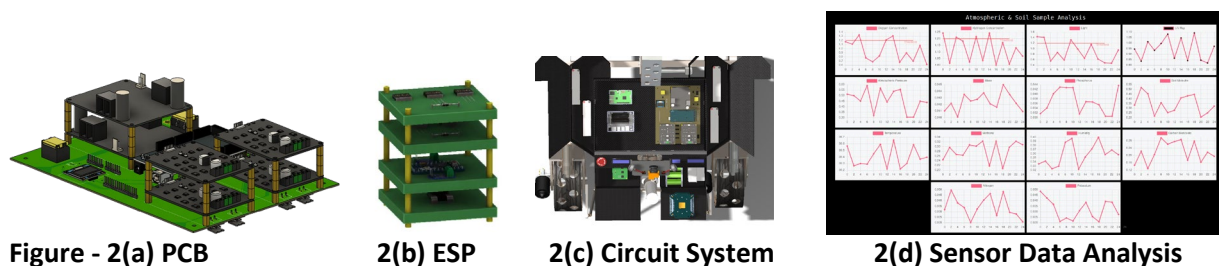
2.2 Network and Vision: Our redesigned and improved network system consists of two distinct but complementary subsystems that differ from the previous year's system. Both FPV and IP camera systems now use a 5.8 GHz system and are powered separately from the main rover system. Two high-end 5.8GHz routers are used in a peer-to-peer network architecture for reliable data transfer between the rover and the base station. The rover is equipped with a 13 dBi omnidirectional antenna (AMO-5G13) paired with a router (Rocket 5ac Lite), while a 19 dBi 120° directional sector antenna (AM-5G19-120) is used at the base station and can rotate up to 360 degrees for coverage of over 1 km. The rover's computing unit is connected to the router using a PoE network switch, which also powers two strategically positioned IP cameras. However, lag and interference issues may arise with longer ranges, hence the use of a 5.8GHz analog FPV system with 2.1mm adjustable lenses. There are four analog cameras equipped with 600 mW transmitter and receiver, and circular polarized mushroom antennas. Two of the FPV cameras can be accessed simultaneously, while the others can be switched through a digital switcher with a pan mechanism. The FPV system is powered using a 12V DC-DC power source, and the PoE switch and routers are powered using a 48V DC-DC power source and PoE converter.

2.3 Control System: This year, we will be controlling the rover from a portable base station that will be operated using two laptops and two displays for camera vision with a keyboard and custom GUI.

2.3.1 Electronics and Power Distribution: The electronics system underwent redesigns; to better serve the different functions of the rover. The power system consists of a 4P4S Lithium-ion Battery (BL18650) that provides 14.6 V and 10 Ah of power for the router and a Lead Acid battery that generates 24V and 9 Ah of power for the rest. The estimated runtime of our rover is 1 hour. Our entire circuit is built on PCB as shown in **figure 2(a)** and a separate slot is allocated for the EPS (electric protection system) as shown in **figure 2(b)**. The custom PCB has Cytron motor drivers stacked on top of one, giving a clean look that makes the circuit easier to debug. A custom kill switch controlled by a 12 V relay with Reverse Polarity protection using MOSFET is implemented. EPS utilizes a current sensor (ACS758LCB) and voltage sensor (B25) to measure power consumption and remaining battery capacity. It protects the PCB from voltage fluctuations, and sensors in EPS send the data to the base station using an Arduino Nano. Dual Channel Cytron 10A DC Motor Driver controls the actuators of the arm, reassuring EMF protection and reliable power conversion. The Single Channel Cytron 30A Bi-Directional DC Motor Driver controls the wheels. A manual & debug board will also be there to aid in troubleshooting the driver board's issues and enable onboard manual control of the rover. In addition, the system features a controller board that interconnects the entire system and incorporates developer boards such as Arduino, Raspberry Pi, and Jetson Nano. The full circuit system can be as seen in **figure 2(c)**.

2.3.2 Software: Using the Robot Operating System (ROS) Noetic on Ubuntu 20.04 LTS, we are controlling the rover with Raspberry Pi 4 as our main control architecture. Through the use of ROS network configuration, we have executed the master node on the base station. The rover receives orders from the base station from a subscriber ROS node, and the commands are delivered to the rover by a publisher ROS node from the base station. Gamepad and Microsoft Flight Controller is used for more ergonomic

and precise maneuvering in order to control the rover. We have implemented ROS for primary communication and a ROS bridge to refine the communication. We have also developed a new robust messaging format that implements error checking and correction, which helps in bug decoding from the base station. We are improving our arms control and movement by implementing inverse kinematics. Our graphical user interface (GUI) is maintained using a website interface to help in swift control of the rover and analyses critical information. The web interface can be accessed from any device such as mobile, laptop, tablet in the local network. Different missions have different tabs in the GUI, and it can calculate various joints' angles on the arm, regulate the speed and check the circuit connection while in operation. We can check the voltage, maintain orientation, inspect the 3D presentation of the rover and check the camera feed area using the GUI. Versatile sensor data streaming format developed for easily managing and debugging various sensors on board the rover. **Figure 2(d)** shows how the sensor data is shown in the GUI. We have implemented a lightweight offline map solution for bird's eye overview of the rover operation during both autonomous missions and manual control. A/B testing for the user-interface and experience is being done actively to make the GUI more accessible and intuitive for any driver. We are working on a dashboard for the rover's GUI where we can see specific real-time information of all the sub-systems for easy diagnosis. A ROS-Gazebo 3D simulation of our rover has been created in order to try and test different algorithms and control modules that are in development or that have already been developed. These simulations have been developed using exact measurements. Using Python socket programming- we established a supplementary control system, and the socket system can access the GUI separately from the ROS system.



3. Approach to Competition Mission: We conducted trial tasks within the given time frame in locations that resembled the competition grounds to carry out the tasks in URC.

3.1 Extreme Retrieval and Delivery Mission: Our rover has a modified rocker-bogie suspension and a sturdy chassis, making it ideal for extreme retrieval and delivery missions. The expandable rover body ensures stability during vertical drops. Our newly modified wheels, equipped with rubber grippers, are designed to provide increased traction, enabling the rover to traverse rocky, uneven, and other similarly challenging terrains. The rover can easily travel up to 1km, and the suspension system can handle sudden jolts while moving through rough surfaces. The 360-degree arm with a three-finger claw is robust enough to lift heavy objects like equipment, screws, and toolboxes with extreme precision. The GUI contains the orientation and the live location of the rover for better spatial perception of the surrounding. There is exact PWM control available to the driver to allow them to have more control over the rover's speed and operations. Using the YOLOv5 algorithm, we have trained an AI model for classifying mechanical tools such as hammers, toolboxes, screwdrivers, and other equipment. We utilized OpenCV and a sharp distance sensor to find the distance between the tool and the rover, helping the driver to pick up efficiently.

3.2 Equipment Servicing Mission: This year, we have used a three-finger claw which gives us better control and precision, improving our performance in the task. Inverse kinematics makes it easier to get accurate movements with the feedback end effector with angle measurements. A laser pointer has been strategically mounted on the arm which helps the driver to easily control and pick objects. With the help

of this mission, we assist the astronauts in the detection of equipment like Toggle Switch, Joystick, Keyboard, and First Aid Box, and instruct the driver for the mission accordingly. Light AI models are used in Jetson Nano for classification and detection.

3.3 Autonomous Traversal Mission: NVIDIA Jetson Nano processes the Autonomous Navigation System while connected to an Arduino Mega through serial communication. Arduino Mega is connected to the peripheral devices. The autonomous navigation is built on top of ROS. Here ROS is being used to communicate different sets of data between to and from nodes. Our main navigation node subscribes to topics regarding GNSS, IMU and ARuCo tag information and uses this data in the algorithm. We are using three sensors this year for our navigation system. Military-grade IMU, WitMotion HWT905 helps to position the rover in the direction of the individual waypoints and gets the compass data for the rover. To find the precise location, we have used two highly accurate SparkFun GPS-RTK2 - one in the base and one in the rover. The base RTK helps to correct the GNSS data and provide us with centimeter level accuracy. To detect any obstacles on its way, we use an RPLiDar A1 to create a map of the surrounding environment within a short distance, and an Intel RealSense D435i depth camera is used within a large area. Both sensors find the shortest path for the rover to reach its waypoints. We detect the AR-tag using the OpenCV ArUco library, and the center of the AR tag is measured using a real sense depth camera. We have developed a custom algorithm that helps the rover to detect and cross the checkpoints more efficiently. The rover rotates at 90-degree angles on each side to find the AR tags. Once detected, it can traverse to the GNSS coordinates autonomously to retrieve the next AR tag data. Rover turns on red LED lights to indicate the start autonomous mode and blinks green LED when the rover reaches a checkpoint. For future development in autonomous missions, we will use a tactical grade dual GNSS INS, SBG Ellipse-D to find highly accurate positional data with drones.

4. Safety Measures: A “kill switch” mechanism implemented by Power MOSFET ensures the rover’s safety compelling the systems to shut down immediately during an emergency. A reverse polarity protection implemented using pMOS. The Cytron motor drivers have back emf protection that keeps the circuit safe from the back emf of the motors.

5. Testing and Operation: Mongol-Tori follows a three-step testing procedure, to make sure that the rover is functioning properly.

a. Sub-System Testing: From 10/08/2022 to 25/12/2022, we completed the subsystem testing, the wheels, suspensions, chassis, sample collector, and science box were designed in the Solidworks program and evaluated through a simulation. After that, the designs were applied and tested. The mechanical structure was tested from 2/2/2023 to 1/3/2023 alongside the network setup which was tested from 14/2/2023 to 1/3/2023. The GUI which is adaptable to any platform was assessed on 30/12/2022.

b. Complete System Testing: Since the URC feedback from the previous year, the rover underwent various modifications this year, with several changes in each subsystem. Therefore after the final assembly, we have a complete rover body and robust AI. The system testing and development of the full rover has been completed within 14/2/2023 and a full rover with full setup has been tested on 2/3/2023.

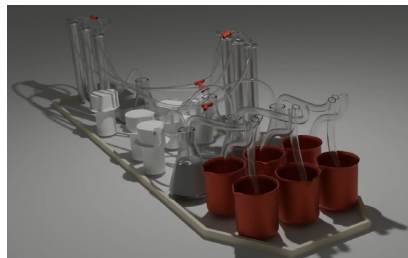
c. Operational Testing: The operational testings were started on 13/2/2023, with autonomous testing carried out on 17/02/2023 at a site with full GPS accessibility. On 19/02/2023, extreme retrieval tasks were conducted and on 23/02/2023, a custom setup has been used to accomplish a variety of equipment service tasks, such as turning a switch on and off, turning knobs, collecting cache boxes, undoing latches, and so on. To identify indications of life, execute microbial biomass testing, capillary water flow testing, and amino acid testing among other things the science module is designed. The rover was intended to gather soil samples utilizing the sample collector on 28/02/2023 to perform the onboard testing and science task.

Science Task: Our main objectives are to investigate the geology and climate of Mars and run experiments to see if life has ever existed there. This will assist astronauts on upcoming missions and help them get ready for space travel. A new, redesigned science module is developed, which ensures all the substances are put more efficiently in distinct compartments. A newly designed science box for the science module is shown in **figures 3(a) and 3(b)**. The newly designed box takes up less space and is designed to conduct onboard science experiments. We are also introducing a PCB as shown in **figure 3(c)** for the science module. This will make the control of the science box easy, and help us conduct tests simultaneously. A science-specific graphical user interface (GUI) is created to help us analyze and send the data we've gathered to the control system. Over the past years, our science module has turned into an autonomous planet analyser. For the science task, we have four kinds of tests divided into two categories, atmospheric analysis, and soil analysis. These are:

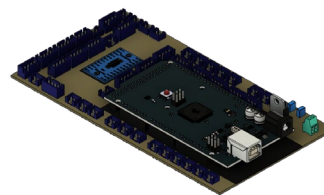
1. Chemical test- Ninhydrin Test, which works on the concept of finding the protein in the soil, will help us determine whether or not there is any potential for life. The nichrome wire gently heats the test tubes and ninhydrin is added. It changes the colour of the soil if an amino acid is present. The iodine Test, allows us to identify the presence of starch in the soil sample. This test's primary objective is to determine whether the soil sample contains plant cells.
2. Physical analysis of soil- Biomass test: This test aids in determining the level of organic compounds present in the soil sample. The measurement for Biomass = $(\text{weight difference}/\text{previous weight}) \times 100$. When we gently heat the soil using a nichrome wire, the load cell detects weight variation for the Biomass test. Using the water flow capillary test, we can determine the soil sample's physical state. It provides a measurement of the amount of water ascending above the surface. The soil absorbs water and the moisture sensor will detect the absorbance level.
3. Probe-based soil analysis- With the help of our NPK test, we can determine if the soil contains any nitrogen, phosphorus, or potassium. These will enable us to determine if the soil contains any living organisms and the data sent by the PCB gets plotted on the graph in the GUI.
4. For atmospheric analysis, we have a sensor module as shown in **figure 3(d)** that works as an automatic onboard analyzer. The box contains a UV Index Sensor (UV GUVA-S12SD), an Air Temperature Sensor (DHT22), an Air Humidity Sensor (DHT22), Sensor-MQ2 (LPG, CO, Smoke, Methane, and Alcohol detection), Sensor-MQ135 (Air Quality Monitoring, detects methane and carbon dioxide. Sensitive to Methane and Carbon dioxide), H₂ Sensor (MQ8), Light Intensity Measurement Sensor (LDR), Sensor BMP-280 (Barometric pressure sensor). Along with data readings for detection, the sensors transmit environmental data to the base point. We can examine the planetary atmosphere, climate, soil content, and landscape formation using sensor data. It aids in understanding both the history and the present of the planet. Additionally, this year we added AI to our science curriculum. The AI model groups and categorizes the rocks based on the microscopic camera's examination of the rocks' texture. The YOLOv5 algorithm built on PyTorch and convolutional neural networks is used for rock classification. Our system will be able to identify the historical traces of life, predict the likelihood of life in the future, and assess changes in the Martian environment by carrying out all these studies.



Figure- 3(a) Box Outside



3(b) Box Inside



3(c) Science PCB



3(d) Sensor Box